

3.1.2 KH₂PO₄, Potassium Dihydrogen Phosphate (KDP)

Negative uniaxial crystal: $n_o > n_e$;

Point group: $\bar{4}2m$;

Mass density: 2.3383 g/cm³ at 293 K [3.59];

Mohs hardness: 2.5;

Transparency range at “0” transmittance level: 0.174 – 1.57 μm [3.60, 59];

Transparency range at 0.5 transmittance level for a 0.8 cm long crystal:

0.178 – 1.45 μm [3.60, 59];

Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
0.212	0.2	3.61	
0.25725	0.01–0.2	3.62	e – wave, $\perp c$
	0.007	3.63	e – wave, $\perp c$
0.3–1.15	< 0.07	3.64	
0.3513	0.003	3.65	e – wave, $\perp c$
0.5145	0.00005	3.62	o – wave
0.5265	0.01	3.66	o – wave
0.94	0.01	3.67	
1.053	0.05	3.66	o – wave
1.054	0.058	3.65	o – wave
	0.02	3.65	e – wave, $\perp c$
1.22	0.1	3.68	o – wave
1.3152	0.3	3.69	
1.32	0.1	3.68	e – wave, $\perp c$

Two-photon absorption coefficient β :

λ [μm]	$\beta \times 10^{13}$ [m/W]	Ref.	Note
0.216	60 ± 5	3.70	
0.2661	27 ± 8.1	3.71	$\theta = 41^\circ, \phi = 45^\circ$
	40–80	3.72	
0.3547	0.59 ± 0.21	3.71	e – wave, $\perp c$

Experimental values of refractive indices at $T = 298$ K [3.73]:

λ [μm]	n_o	n_e	λ [μm]	n_o	n_e
0.2138560	1.60177	1.54615	0.2980628	1.54618	1.49824
0.2288018	1.58546		0.3021499	1.54433	1.49708
0.2446905	1.57228		0.3035781		1.49667
0.2464068	1.57105		0.3125663	1.54117	1.49434
0.2536519	1.56631	1.51586	0.3131545	1.54098	1.49419
0.2800869	1.55263	1.50416	0.3341478		1.48954

0.3650146	1.52932	1.48432	0.5769580	1.50987	
0.3654833	1.52923	1.48423	0.5790654	1.50977	1.46856
0.3662878	1.52909	1.48409	0.6328160	1.50737	1.46685
0.3906410		1.48089	1.0139750	1.49535	1.46041
0.4046561	1.52341	1.47927	1.1287040	1.49205	1.45917
0.4077811	1.52301	1.47898	1.1522760	1.49135	1.45893
0.4358350	1.51990	1.47640	1.3570700	1.48455	
0.4916036		1.47254	1.5231000		1.45521
0.5460740	1.51152	1.46982	1.5295250		1.45512

Temperature derivative of refractive indices [3.74]:

λ [μm]	$dn_o/dT \times 10^5$ [K^{-1}]	$dn_e/dT \times 10^5$ [K^{-1}]
0.405	-3.27	-3.15
0.436	-3.27	-2.88
0.546	-3.28	-2.90
0.578	-3.25	-2.87
0.633	-3.94	-2.54

Temperature dependences of refractive indices upon cooling from room temperature to T [K].

for the spectral range 0.365 – 0.690 μm [3.75]:

$$n_o(T) = n_o(298) + 0.402 \times 10^{-4} \{ [n_o(298)]^2 - 1.432 \} (298 - T) ;$$

$$n_e(T) = n_e(298) + 0.221 \times 10^{-4} \{ [n_e(298)]^2 - 1.105 \} (298 - T) ;$$

for the spectral range 0.436 – 0.589 μm [3.76]:

$$n_o(T) = n_o(300) + 10^{-4} (143.3 - 0.618T + 4.81 \times 10^{-4} T^2) ,$$

$$n_e(T) = n_e(300) + 10^{-4} (153.3 - 0.969T + 1.57 \times 10^{-3} T^2) .$$

Experimental values of phase-matching angle ($T = 293$ K) and comparison between different sets of dispersion relations:

Interacting wavelengths [μm]	θ_{exp} [deg]	θ_{theor} [deg]		
		[3.73]	[3.77]	[3.78]K
SHG, o + o ⇒ e				
0.517 ⇒ 0.2585	90 [3.74]	no pm	no pm	73.6
0.6576 ⇒ 0.3288	53.6 [3.69]	53.6	53.6	53.2
0.6943 ⇒ 0.34715	50.4 [3.79]	50.6	50.6	50.4
0.8707 ⇒ 0.43535	42.4 [3.80]	42.8	42.7	42.8
1.06 ⇒ 0.53	41 [3.81]	41.2	41.0	40.9
	41 [3.82]			

1.3152 $\Rightarrow$ 0.6576	44.3 [3.69]	44.6	44.7	44.1
SFG, o + o $\Rightarrow$ e				
1.415 + 0.22027 $\Rightarrow$				
$\Rightarrow$ 0.1906	88.7 [3.83]	83.7	83.6	54.3
1.3648 + 0.6943 $\Rightarrow$				
$\Rightarrow$ 0.46019	40.9 [3.80]	41.7	41.7	41.6
1.3152 + 0.6576 $\Rightarrow$				
$\Rightarrow$ 0.4384	42.2 [3.69]	42.1	42.1	42.0
1.0642 + 0.2707 $\Rightarrow$				
$\Rightarrow$ 0.21581	87.6 [3.84]	87.5	87.3	62.9
1.0642 + 0.5321 $\Rightarrow$				
$\Rightarrow$ 0.35473	47.3 [3.85]	47.3	47.3	47.1
1.06 + 0.53 $\Rightarrow$				
$\Rightarrow$ 0.35333	47.5 [3.82]	47.4	47.4	47.3
0.6576 + 0.4384 $\Rightarrow$				
$\Rightarrow$ 0.26304	74 [3.86]	75.2	75.4	68.6
SHG, e + o $\Rightarrow$ e				
1.3152 $\Rightarrow$ 0.6576	61.4 [3.69]	61.8	61.8	60.7
1.06 $\Rightarrow$ 0.53	59 [3.82]	59.0	58.8	58.6
SFG, e + o $\Rightarrow$ e				
1.0642 + 0.5321 $\Rightarrow$				
$\Rightarrow$ 0.35473	58.3 [3.85]	58.2	58.3	57.9
1.06 + 0.53 $\Rightarrow$				
$\Rightarrow$ 0.35333	59.3 [3.82]	58.5	58.5	58.1

Note: The other sets of dispersion relations from [3.74] and [3, 78]E show worse agreement with the experiment.

[3.78]K $\Rightarrow$ see [3.78], data of Kirby et al.;

[3.78]E $\Rightarrow$ see [3.78], data of Eimerl.

Experimental values of NCPM temperature:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	Ref.
SHG, o + o $\Rightarrow$ e		
0.5145 $\Rightarrow$ 0.25725	-13.7	3.63
	-11	3.62
0.517 $\Rightarrow$ 0.2585	20	3.74
0.5321 $\Rightarrow$ 0.26605	177	3.87
	177	3.88
SFG, o + o $\Rightarrow$ e		
1.06 + 0.265 $\Rightarrow$ 0.212	-70	3.61
1.0642 + 0.26605 $\Rightarrow$ 0.21284	-40	3.89
	-35	3.90

Best set of dispersion relations (λ in μm , $T = 20^\circ\text{C}$) [3.74] :

$$n_o^2 = 2.259276 + \frac{13.00522\lambda^2}{\lambda^2 - 400} + \frac{0.01008956}{\lambda^2 - (77.26408)^{-1}} ,$$

$$n_e^2 = 2.132668 + \frac{3.2279924\lambda^2}{\lambda^2 - 400} + \frac{0.008637494}{\lambda^2 - (81.42631)^{-1}} .$$

Temperature-dependent Sellmeier equations (λ in μm , T in K) [3.77] :

$$n_o^2 = (1.44896 + 3.185 \times 10^{-5} T) + \frac{(0.84181 - 1.4114 \times 10^{-4} T)\lambda^2}{\lambda^2 - (0.0128 - 2.13 \times 10^{-7} T)}$$

$$+ \frac{(0.90793 + 5.75 \times 10^{-7} T)\lambda^2}{\lambda^2 - 30} ,$$

$$n_e^2 = (1.42691 - 1.152 \times 10^{-5} T) + \frac{(0.72722 - 6.139 \times 10^{-5} T)\lambda^2}{\lambda^2 - (0.01213 + 3.104 \times 10^{-7} T)}$$

$$+ \frac{(0.22543 - 1.98 \times 10^{-7} T)\lambda^2}{\lambda^2 - 30} .$$

Calculated values of phase-matching and “walk-off” angles:

Interacting wavelengths [μm]	θ_{pm} [deg]	ρ_1 [deg]	ρ_3 [deg]
SHG, o + o $\Rightarrow$ e			
0.5321 $\Rightarrow$ 0.26605	76.60		0.808
0.5782 $\Rightarrow$ 0.2891	64.03		1.391
0.6328 $\Rightarrow$ 0.3164	56.15		1.611
0.6594 $\Rightarrow$ 0.3297	53.43		1.657
0.6943 $\Rightarrow$ 0.34715	50.55		1.687
1.0642 $\Rightarrow$ 0.5321	41.21		1.603
1.3188 $\Rightarrow$ 0.6594	44.70		1.549
SFG, o + o $\Rightarrow$ e			
0.5782 + 0.5105 $\Rightarrow$ 0.27112	72.46		1.025
1.0642 + 0.5321 $\Rightarrow$ 0.35473	47.28		1.712
1.3188 + 0.6594 $\Rightarrow$ 0.4396	42.05		1.657
SHG, e + o $\Rightarrow$ e			
1.0642 $\Rightarrow$ 0.5321	58.98	1.149	1.404
1.3188 $\Rightarrow$ 0.6594	61.85	0.922	1.269
SFG, e + o $\Rightarrow$ e			
1.0642 + 0.5321 $\Rightarrow$ 0.35473	58.23	1.166	1.521
1.3188 + 0.6594 $\Rightarrow$ 0.4396	49.42	1.104	1.634

Calculated values of inverse group-velocity mismatch for SHG process in KDP:

Interacting wavelengths [μm]	θ_{pm} [deg]	β [fs/mm]
SHG, $o + o \Rightarrow e$		
$1.2 \Rightarrow 0.6$	42.45	42
$1.1 \Rightarrow 0.55$	41.38	17
$1.0 \Rightarrow 0.5$	41.22	9
$0.9 \Rightarrow 0.45$	42.24	40
$0.8 \Rightarrow 0.4$	44.91	77
$0.7 \Rightarrow 0.35$	50.14	128
$0.6 \Rightarrow 0.3$	60.40	208
SHG, $e + o \Rightarrow e$		
$1.2 \Rightarrow 0.6$	59.54	89
$1.1 \Rightarrow 0.55$	58.87	67
$1.0 \Rightarrow 0.5$	59.75	89
$0.9 \Rightarrow 0.45$	62.97	118
$0.8 \Rightarrow 0.4$	70.71	158

Experimental values of internal angular and temperature bandwidths:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	θ_{pm} [deg]	$\Delta\theta^{\text{int}}$ [deg]	ΔT [$^{\circ}\text{C}$]	Ref.
SHG, $o + o \Rightarrow e$					
$1.1523 \Rightarrow 0.57615$	20	41	0.074		3.91
$1.0642 \Rightarrow 0.5321$	20	41	0.070		3.92
	25			23	3.93
$1.064 \Rightarrow 0.532$	20	41	0.069		3.94
$1.06 \Rightarrow 0.53$	20	41	0.063		3.81
$1.054 \Rightarrow 0.527$	25	41	0.060		3.95
$0.5321 \Rightarrow 0.26605$	25			1.7	3.93
	177	90		1.9	3.87
	177	90		2	3.88
$0.53 \Rightarrow 0.265$	20	77	0.059		3.96
	20	77	0.066		3.97
SFG, $o + o \Rightarrow e$					
$1.0642 + 0.5321 \Rightarrow$ $\Rightarrow 0.35473$	25			5.5	3.93
$1.054 + 0.527 \Rightarrow$ $\Rightarrow 0.35133$	25	48	0.046		3.95
SHG, $e + o \Rightarrow e$					
$1.0642 \Rightarrow 0.5321$	25			18.3	3.93
$1.06 \Rightarrow 0.53$	20	59	0.129		3.96

1.054 $\Rightarrow$ 0.527	25	59	0.126	3.95
SFG, e + o $\Rightarrow$ e				
1.0642 + 0.5321 $\Rightarrow$ $\Rightarrow$ 0.35473	25		5.2	3.93
1.06 + 0.53 $\Rightarrow$ $\Rightarrow$ 0.35333	20	59	0.062	3.97
1.054 + 0.527 $\Rightarrow$ $\Rightarrow$ 0.35133	25	59	0.059	3.95

Experimental values of spectral bandwidth:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	θ_{pm} [deg]	$\Delta\nu$ [cm^{-1}]	Ref.
SHG, o + o $\Rightarrow$ e				
1.06 $\Rightarrow$ 0.53	20	41	178	3.81
0.53 $\Rightarrow$ 0.265	20	77	1.2	3.96
SHG, e + o $\Rightarrow$ e				
1.06 $\Rightarrow$ 0.53	20	59	101.5	3.96

Temperature variation of phase-matching angle:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	θ_{pm} [deg]	$d\theta_{\text{pm}}/dT$ [deg/K]	Ref.
SHG, o + o $\Rightarrow$ e				
1.0642 $\Rightarrow$ 0.5321	25		0.0028	3.93
1.054 $\Rightarrow$ 0.527	25	41	0.0046	3.95
0.5321 $\Rightarrow$ 0.26605	25		0.0382	3.93
SFG, o + o $\Rightarrow$ e				
1.0642 + 0.5321 $\Rightarrow$ 0.35473	25		0.0073	3.93
1.054 + 0.527 $\Rightarrow$ 0.35133	25	59	0.0046	3.95
SHG, e + o $\Rightarrow$ e				
1.0642 $\Rightarrow$ 0.5321	25	59	0.0069	3.98
	25		0.0069	3.93
1.06 $\Rightarrow$ 0.53	20	59	0.0057	3.96
1.054 $\Rightarrow$ 0.527	25	59	0.0086	3.95
	20	59	0.0069	3.65
SFG, e + o $\Rightarrow$ e				
1.0642 + 0.5321 $\Rightarrow$ 0.35473	25	58	0.0106	3.98
	25		0.0117	3.93
1.054 + 0.527 $\Rightarrow$ 0.35133	25	59	0.0152	3.95
	20	59	0.0075	3.65

Temperature tuning of noncritical SHG [3.74]:

Interacting wavelengths [μm]	$d\lambda_1/dT$ [nm/K]
SHG, $o + o \Rightarrow e$ 0.517 $\Rightarrow$ 0.2585	0.048

Temperature variation of birefringence for noncritical SHG process:

Interacting wavelengths [μm]	$d(n_2^e - n_1^o)/dT \times 10^5$ [K^{-1}]	Ref.
0.5145 $\Rightarrow$ 0.25725	1.745	3.99
0.5321 $\Rightarrow$ 0.26605	1.2	3.87

Effective nonlinearity expressions in the phase-matching direction [3.100]:

$$d_{\text{ooe}} = d_{36} \sin \theta \sin 2\phi ,$$

$$d_{\text{eoe}} = d_{\text{oeo}} = d_{36} \sin 2\theta \cos 2\phi .$$

Nonlinear coefficient [3.37]:

$$d_{36}(1.064 \mu\text{m}) = 0.39 \text{ pm/V} ,$$

Laser-induced bulk-damage threshold:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m^2]	Ref.
0.52	330	2	3.101
0.5265	20	30	3.66
	0.6	90	3.66
0.527	0.5	> 140	3.102
0.53	0.2	170	3.103
	0.005	100 00(?)	3.104
0.5321	0.6	> 80	3.72
	0.03	300	3.105
0.596	330	2.4	3.101
	20	30	3.101
0.6943	20	> 4	3.106
1.053	25	40	3.66
	1	180	3.66
	1	200	3.107
1.054	0.14	> 70	3.108
1.06	60	2	3.109
	12–25	2.5	3.81
	0.5	> 30	3.110
	0.2	230	3.103
1.064	20	3–6	3.111
	1.3	80	3.33

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m^2]	Ref.
1.064	1	30–70	3.111
	1	50	3.112
	0.1	70	3.1

Thermal conductivity coefficient [3.59]:

T[K]	κ [W/mK], $\parallel c$	κ [W/mK], $\perp c$
302	1.21	
319		1.34

3.1.3 KD_2PO_4 , Deuterated Potassium Dihydrogen Phosphate (DKDP)

Negative uniaxial crystal: $n_o > n_e$;

Point group: $\bar{4}2\text{m}$;

Mass density: $2.355 \text{ g}/\text{cm}^3$;

Mohs hardness: 2.5;

Transparency range at “0” transmittance level: $0.2 - 2.1 \mu\text{m}$ [3.113, 114];

Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
0.266	0.035	3.115	
0.5321	0.004–0.005	3.116	98–99% deuteration
0.82–1.21	< 0.015	3.67	
0.94	0.005	3.67	
1.0642	0.004–0.005	3.116	98–99% deuteration
1.315	0.025	3.117	
1.57	0.1	3.68	o – wave, 95% deuteration
1.74	0.1	3.68	e – wave, 95% deuteration

Two-photon absorption coefficient β :

λ [μm]	$\beta \times 10^{13}$ [m/W]	Ref.	Note
0.2661	2.0 ± 1.0	3.118	
	2.7 ± 0.7	3.115	
0.3547	0.54 ± 0.19	3.71	e – wave, $\perp c$

Experimental values of refractive indices at $T = 298 \text{ K}$ [3.95]:

λ [μm]	n_o	n_e
0.4047	1.5189	1.4776
0.4078	1.5185	1.4772

λ [μm]	n_o	n_e
0.4358	1.5155	1.4747
0.4916	1.5111	1.4710
0.5461	1.5079	1.4683
0.5779	1.5063	1.4670
0.6234	1.5044	1.4656
0.6907	1.5022	1.4639

Temperature derivative of refractive indices [3.74]:

λ [μm]	$dn_o/dT \times 10^5$ [K^{-1}]	$dn_e/dT \times 10^5$ [K^{-1}]
0.405	-3.00	-1.86
0.436	-3.37	-2.13
0.546	-2.99	-1.95
0.578	-3.00	-2.52
0.633	-3.16	-2.03

Temperature dependences of refractive indices upon cooling from room temperature to T [K]

for the spectral range 0.365 – 0.690 μm [3.75] :

$$n_o(T) = n_o(298) + 0.228 \times 10^{-4} \{ [n_o(298)]^2 - 1.047 \} (298 - T) ;$$

$$n_e(T) = n_e(298) + 0.955 \times 10^{-5} [n_e(298)]^2 (298 - T) ;$$

for the spectral range 0.436 – 0.589 μm [3.76]:

$$n_o(T) = n_o(300) + 10^{-4} (85.2 - 0.0695 T - 7.25 \times 10^{-4} T^2) ,$$

$$n_e(T) = n_e(300) + 10^{-4} (21.8 - 0.445 T - 1.24 \times 10^{-3} T^2) .$$

Experimental values of phase-matching angle ($T = 293$ K) and comparison between different sets of dispersion relations:

Interacting wavelengths [μm]	θ_{exp} [deg]	θ_{theor} [deg]		
		[3.77]	[3.78]K	[3.78]E
SHG, o + o ⇒ e				
0.530 ⇒ 0.265	90 [3.119]	no pm	no pm	87.4
0.6943 ⇒ 0.34715	52 [3.79]	50.6	50.9	51.0
1.062 ⇒ 0.531	37.1 [3.120]	38.6	36.6	36.6
SHG, e + o ⇒ e				
1.3152 ⇒ 0.6576	51.3 [3.69]	63.2	51.7	49.4

Note: The set of dispersion relations from [3.74] shows worse agreement with the experiment.

[3.78]K $\Rightarrow$ see [3.78], data of Kirby et al.;

[3.78]E $\Rightarrow$ see [3.78], data of *Eimerl*.

Experimental values of NCPM temperature:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	Ref.	Note
SHG, $o + o \Rightarrow e$			
$0.528 \Rightarrow 0.264$	-30	3.119	
$0.5321 \Rightarrow 0.26605$	42	3.89	99% deuteration
	45	3.87	95% deuteration
	46	3.90	99% deuteration
	49.8	3.121	> 95% deuteration
	60.8	3.122	90% deuteration
$0.536 \Rightarrow 0.268$	100	3.119	

Best set of dispersion relations (λ in μm , $T = 20^{\circ}\text{C}$) [3.78]K :

$$n_o^2 = 2.240921 + \frac{2.246956\lambda^2}{\lambda^2 - (11.26591)^2} + \frac{0.009676}{\lambda^2 - (0.124981)^2} ,$$

$$n_e^2 = 2.126019 + \frac{0.784404\lambda^2}{\lambda^2 - (11.10871)^2} + \frac{0.008578}{\lambda^2 - (0.109505)^2} .$$

Temperature-dependent Sellmeier equations (λ in μm , T in K) [3.77] :

$$n_o^2 = (1.55934 + 3.3935 \times 10^{-4} T) + \frac{(0.71098 - 4.1655 \times 10^{-4} T)\lambda^2}{\lambda^2 - (0.01407 + 6.4904 \times 10^{-6} T)} + \frac{(0.67671 + 4.8281 \times 10^{-5} T)\lambda^2}{\lambda^2 - 30} ,$$

$$n_e^2 = (1.68647 + 3.43 \times 10^{-6} T) + \frac{(0.46629 - 6.26 \times 10^{-5} T)\lambda^2}{\lambda^2 - (0.01663 + 1.3626 \times 10^{-6} T)} + \frac{(0.59614 + 2.41 \times 10^{-7} T)\lambda^2}{\lambda^2 - 30} .$$

Calculated values of phase-matching and “walk-off” angles:

Interacting wavelengths [μm]	θ_{pm} [deg]	ρ_1 [deg]	ρ_3 [deg]
SHG, $o + o \Rightarrow e$			
$0.5321 \Rightarrow 0.26605$	86.20		0.225
$0.5782 \Rightarrow 0.2891$	66.87		1.197
$0.6328 \Rightarrow 0.3164$	57.53		1.467
$0.6594 \Rightarrow 0.3297$	54.31		1.522
$1.6943 \Rightarrow 0.34715$	50.86		1.558
$1.0642 \Rightarrow 0.5321$	36.60		1.450
$1.3188 \Rightarrow 0.6594$	36.36		1.412

SFG, $o + o \Rightarrow e$			
$0.5782 + 0.5105 \Rightarrow 0.27112$	77.88		0.595
$1.0642 + 0.5321 \Rightarrow 0.35473$	46.82		1.580
$1.3188 + 0.6594 \Rightarrow 0.4396$	39.18		1.515
SHG, $e + o \Rightarrow e$			
$1.0642 \Rightarrow 0.5321$	53.47	1.286	1.427
$1.3188 \Rightarrow 0.6594$	51.70	1.222	1.420
SFG, $e + o \Rightarrow e$			
$1.0642 + 0.5321 \Rightarrow 0.35473$	59.38	1.174	1.378
$1.3188 + 0.6594 \Rightarrow 0.4396$	47.70	1.254	1.527

Calculated values of inverse group-velocity mismatch for SHG process in DKDP:

Interacting wavelengths [μm]	θ_{pm} [deg]	β [fs/mm]
SHG, $o + o \Rightarrow e$		
$1.2 \Rightarrow 0.6$	35.94	< 1
$1.1 \Rightarrow 0.55$	36.28	18
$1.0 \Rightarrow 0.5$	37.47	38
$0.9 \Rightarrow 0.45$	39.79	63
$0.8 \Rightarrow 0.4$	43.75	96
$0.7 \Rightarrow 0.35$	50.37	143
$0.6 \Rightarrow 0.3$	62.54	218
SHG, $e + o \Rightarrow e$		
$1.2 \Rightarrow 0.6$	51.62	55
$1.1 \Rightarrow 0.55$	52.73	71
$1.0 \Rightarrow 0.5$	55.37	92
$0.9 \Rightarrow 0.45$	60.41	120
$0.8 \Rightarrow 0.4$	70.43	159

Experimental values of internal angular and temperature bandwidths:

Interacting wavelengths [μm]	<i>T</i> [°C]	<i>θ</i> _{pm} [deg]	<i>Δθ</i> ^{int} [deg]	<i>ΔT</i> [°C]	Ref.
SHG, o + o ⇒ e					
1.0642 ⇒ 0.5321	20	37	0.081		3.92
0.5321 ⇒ 0.26605	60.8	90		1.8	3.122
	45	90		1.9	3.87
SHG, e + o ⇒ e					
1.0642 ⇒ 0.5321	20	54	0.131		3.123
	20		0.126		3.124
1.06 ⇒ 0.53	20	60	0.143		3.96

Experimental value of spectral bandwidth [3.96]:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	θ_{pm} [deg]	$\Delta\nu$ [cm^{-1}]
SHG, $e + o \Rightarrow e$ $1.06 \Rightarrow 0.53$	20	60	74.8

Temperature variation of phase-matching angle [3.96]:

Interacting wavelengths [μm]	T [$^{\circ}\text{C}$]	θ_{pm} [deg]	$d\theta_{\text{pm}}/dT$ [deg/K]
SHG, $e + o \Rightarrow e$ $1.06 \Rightarrow 0.53$	20	60	0.0063

Temperature tuning of noncritical SHG [3.74]:

Interacting wavelengths [μm]	$d\lambda_1/dT$ [nm/K]
SHG, $o + o \Rightarrow e$ $0.519 \Rightarrow 0.2595$	0.068

Effective nonlinearity in the phase-matching direction [3.100]:

$$d_{\text{oeo}} = d_{36} \sin \theta \sin 2\phi ,$$

$$d_{\text{eoe}} = d_{\text{ooo}} = d_{36} \sin 2\theta \cos 2\phi .$$

Nonlinear coefficient [3.37]:

$$d_{36}(1.064\mu\text{m}) = 0.37 \text{ pm/V} .$$

Laser-induced bulk-damage threshold:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m 2]	Ref.
0.266	0.03	> 100	3.115
0.532	30	> 0.5	3.122
	8	170	3.125
	0.6	> 80	3.72
	0.03	> 80	3.118
0.6	330	3	3.101
1.062	0.007	> 10	3.120
1.064	40	> 2.5	3.122
	18	> 1.0	3.116
	14	80	3.125
	1	60	3.124
	0.25	> 30	3.116
1.315	1	15	3.69

Thermal conductivity coefficient [3.78]:

$$\kappa = 1.86 \text{ Wm/K } (\parallel c) ,$$

$$\kappa = 2.09 \text{ Wm/K } (\perp c) .$$

3.1.4 $\text{NH}_4\text{H}_2\text{PO}_4$, Ammonium Dihydrogen Phosphate (ADP)

Negative uniaxial crystal : $n_o > n_e$;

Point group: $\bar{4}2m$;

Mass density: 1.803 g/cm^3 at 293 K [3.59];

Mohs hardness: 2;

Transparency range at "0" transmittance level: $0.18 - 1.53 \mu\text{m}$ [3.60, 126];

Transparency range at 0.5 transmittance level for a 0.8 cm long crystal:

$0.185 - 1.45 \mu\text{m}$ [3.60, 59]

Linear absorption coefficient α :

$\lambda [\mu\text{m}]$	$\alpha [\text{cm}^{-1}]$	Ref.	Note
0.25725	0.002	3.62	e – wave, $\perp c$
0.265	0.07	3.127	e – wave, $\perp c$
0.266	0.035	3.115	
0.3–1.15	< 0.07	3.64	
0.5145	0.00005	3.62	o – wave, $\perp c$
1.027	0.086	3.67	
1.083	0.208	3.67	
1.144	0.150	3.67	

Two-photon absorption coefficient β :

$\lambda [\mu\text{m}]$	$\beta \times 10^{13} [\text{m/W}]$	Ref.	Note
0.2661	6 ± 1	3.118	
	11 ± 3	3.115	
	24 ± 7	3.71	$\theta = 42^\circ, \phi = 45^\circ$
0.3078	23 ± 5	3.128	
0.3547	0.68 ± 0.24	3.71	e – wave, $\perp c$

Experimental values of refractive indices at $T = 298 \text{ K}$ [3.73, 129]:

$\lambda [\mu\text{m}]$	n_o	n_e	$\lambda [\mu\text{m}]$	n_o	n_e
0.2138560	1.62598	1.56738	0.3021499	1.56270	1.51163
0.2288018	1.60785	1.55138	0.3125663	1.55917	1.50853
0.2536519	1.58688	1.53289	0.3131545	1.55897	1.50832
0.2967278	1.56462	1.51339	0.3341478	1.55300	1.50313

Thermal conductivity coefficient:

κ [W/mK], $\parallel c$	κ [W/mK], $\perp c$	Ref.
0.8	0.08	3.148
1.6	1.2	3.58

3.1.6 LiIO₃, Lithium Iodate

Negative uniaxial crystal: $n_o > n_e$;

Point group: 6;

Mass density: 4.49 g/cm³ [3.182];

Mohs hardness: 3.5 – 4.0;

Transparency range at “0” transmittance level: 0.28 – 6 μm [3.183, 184];Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
0.34715	0.1	3.185	$\parallel c$
	0.3	3.185	e – wave, $\perp c$
0.5145	0.0024	3.186	$\parallel c$
	0.0025	3.186	e – wave, $\perp c$
0.5321	0.3	3.187	e – wave
0.5422	0.37	3.184	
0.6594	0.0007–0.0023	3.186	$\parallel c$
	0.0006–0.0017	3.186	e – wave, $\perp c$
1.0642	0.1	3.187	o – wave
	0.25	3.187	e – wave
	< 0.0002	3.186	$\parallel c$
	0.0008	3.186	e – wave, $\perp c$
1.0845	0.06	3.184	
1.315	0.0005	3.182	
1.3188	0.0008–0.0036	3.186	$\parallel c$
	0.0007–0.0010	3.186	e – wave, $\perp c$

Two-photon absorption coefficient at $\lambda = 0.532 \mu\text{m}$: $\beta < 4 \times 10^{-12} \text{ m/W}$ [3.188].

Experimental values of refractive indices:

λ [μm]	n_o	n_e	Ref.	λ [μm]	n_o	n_e	Ref.
0.3547	1.9822	1.8113	3.189	0.3996	1.9464	1.7842	3.189
0.3669	1.9706	1.8026	3.189	0.4047	1.9443	1.7826	3.190
0.3712	1.9671	1.8000	3.189	0.4358	1.9275	1.7702	3.189
0.3795	1.9600	1.7947	3.189	0.4545	1.9184	1.7638	3.191
0.3877	1.9544	1.7905	3.189	0.4579	1.9170	1.7630	3.191

λ [μm]	n_o	n_e	Ref.	λ [μm]	n_o	n_e	Ref.
0.4658	1.9141	1.7611	3.191	0.7000	1.8746	1.7300	3.96
0.4727	1.9122	1.7600	3.191	0.7660	1.8694	1.7261	3.96
0.4765	1.9100	1.7583	3.191	0.8000	1.8673	1.7245	3.96
0.4800	1.9109	1.7579	3.189	0.8630	1.8640	1.7220	3.96
0.4880	1.9083	1.7556	3.191	0.9000	1.8623	1.7207	3.96
0.5017	1.9053	1.7537	3.191	1.0000	1.8587	1.7180	3.96
0.5086	1.9031	1.7514	3.189	1.1000	1.8559	1.7160	3.96
0.5145	1.9012	1.7487	3.191	1.2000	1.8536	1.7143	3.96
0.5320	1.8975	1.7475	3.189	1.3000	1.8517	1.7130	3.96
0.5461	1.8950	1.7455	3.96	1.3674	1.8508	1.7122	3.190
0.5600	1.8921	1.7433	3.189	1.5296	1.8482	1.7101	3.190
0.5791	1.8894	1.7413	3.190	1.6920	1.8464	1.7089	3.190
0.5800	1.8889	1.7403	3.189	1.9701	1.8431	1.7072	3.190
0.5896	1.8875	1.7400	3.190	2.2493	1.8385	1.7050	3.190
0.6000	1.8859	1.7383	3.189	2.5000	1.8378	1.7037	3.192
0.6200	1.8828	1.7361	3.189	3.0000	1.8319	1.7001	3.192
0.6328	1.8815	1.7351	3.192	3.5000	1.8266	1.6971	3.192
0.6438	1.8807	1.7346	3.190	4.0000	1.8140	1.6897	3.192
0.6560	1.8789	1.7332	3.96	5.0000	1.7940	1.6783	3.192

Optical activity at $T = 300$ K :

λ [μm]	ρ [deg/mm]	Ref.	λ [μm]	ρ [deg/mm]	Ref.
0.286	1052.9	3.193	0.429	222.46	3.193
0.290	964.99	3.193	0.448	198.72	3.193
0.295	886.65	3.193	0.470	175.75	3.193
0.299	814.39	3.193	0.492	153.61	3.193
0.304	748.76	3.193	0.520	133.02	3.193
0.310	687.46	3.193	0.546	117.42	3.193
0.317	630.44	3.193	0.551	113.36	3.193
0.324	579.01	3.193	0.600	95.27	3.193
0.331	532.44	3.193	0.628	86.80	3.193
0.339	489.47	3.193	1.084	25.0	3.184
0.347	448.42	3.193	1.1	23.83	3.194
0.355	410.37	3.193	1.6	11.00	3.194
0.363	374.34	3.193	2.1	6.33	3.194
0.374	340.18	3.193	2.6	4.12	3.194
0.386	308.07	3.193	3.1	2.89	3.194
0.399	277.45	3.193	3.6	2.32	3.194
0.412	249.32	3.193			

Temperature derivative of refractive indices [3.186]:

λ [μm]	$dn_o/dT \times 10^5$ [K^{-1}]	$dn_e/dT \times 10^5$ [K^{-1}]
0.5321	-9.64	-8.61
0.6594	-9.49	-8.39
1.0642	-8.93	-7.52
1.3188	-9.44	-8.49

Experimental values of phase-matching angle ($T = 293$ K) and comparison between different sets of dispersion relations:

Interacting wavelengths [μm]	θ_{exp} [deg]	θ_{theor} [deg]		
		[3.195]	[3.192]	[3.196]
SHG, $\text{o} + \text{o} \Rightarrow \text{e}$				
$0.586 \Rightarrow 0.293$	90 [3.195]	87.7	81.0	83.6
$0.5863 \Rightarrow 0.29315$	90 [3.196]	86.9	80.7	83.2
$0.6 \Rightarrow 0.3$	75.6 [3.196]	75.5	73.5	73.7
$0.62 \Rightarrow 0.31$	68.2 [3.196]	68.0	67.0	66.3
$0.6943 \Rightarrow 0.34715$	52 [3.197]	53.5	53.4	52.0
	52 [3.198]			
$1.06 \Rightarrow 0.53$	30 [3.199]	30.2	30.3	29.5
$1.0642 \Rightarrow 0.5321$	30.2 [3.200]	30.1	30.2	29.4
	30.2 [3.92]			
	30 [3.201]			
	30 [3.202]			
$1.0845 \Rightarrow 0.54225$	28.9 [3.184]	29.5	29.6	28.8
$1.1523 \Rightarrow 0.57615$	27.2 [3.184]	27.6	27.7	27.1
$1.3886 \Rightarrow 0.6943$	23.1 [3.203]	23.2	23.2	23.1
$1.746 \Rightarrow 0.873$	20 [3.204]	19.9	19.6	20.3
SFG, $\text{o} + \text{o} \Rightarrow \text{e}$				
$5.33 + 1.32969 \Rightarrow 1.0642$	21 [3.202]	20.1	20.0	22.1
$4.44 + 1.39968 \Rightarrow 1.0642$	20.2 [3.202]	19.4	19.0	21.2
$5.2 + 0.80129 \Rightarrow 0.6943$	19.5 [3.205]	19.6	19.6	20.6
$2.5 + 0.96126 \Rightarrow 0.6943$	21 [3.206]	21.5	21.4	21.7
$5.0 + 0.66251 \Rightarrow 0.585$	20.3 [3.207]	20.3	20.3	20.9
$2.0 + 0.82686 \Rightarrow 0.585$	25.1 [3.207]	25.1	25.1	24.8
$4.16 + 0.61015 \Rightarrow 0.5321$	21.6 [3.208]	21.7	21.7	22.0
$2.66 + 0.66514 \Rightarrow 0.5321$	24.5 [3.208]	24.6	24.6	24.4
$0.946 + 0.5484 \Rightarrow 0.34715$	50 [3.209]	51.0	50.9	49.6
$2.67 + 0.6943 \Rightarrow 0.55102$	24.4 [3.210]	24.1	24.1	23.9
$1.98 + 0.6943 \Rightarrow 0.51405$	27.4 [3.210]	27.6	27.6	27.1
$1.2013 + 0.6943 \Rightarrow 0.44$	35.1 [3.203]	36.2	36.4	35.2
$3.3913 + 0.5145 \Rightarrow 0.44673$	24 [3.211]	25.4	25.4	25.1
$2.38 + 0.4880 \Rightarrow 0.40497$	30.5 [3.212]	31.1	31.1	30.5
$1.0642 + 0.5321 \Rightarrow 0.35473$	47.5 [3.200]	47.8	47.8	46.5

Note: the other sets of dispersion relations from [3.213, 214, 215, 189, 202] show worse agreement with the experiment.

Best set of dispersion relations (λ in μm , $T = 20^\circ\text{C}$) [3.295] (a corrected set from [3.213]):

$$n_o^2 = 3.4132 + \frac{0.0476}{\lambda^2 - 0.0338} - 0.0077 \lambda^2,$$

$$n_e^2 = 2.9211 + \frac{0.0346}{\lambda^2 - 0.0320} - 0.0042 \lambda^2.$$

Calculated values of phase-matching and “walk-off” angles:

Interacting wavelengths [μm]	θ_{pm} [deg]	ρ_3 [deg]
SHG, $o + o \Rightarrow e$		
$0.6328 \Rightarrow 0.3164$	64.52	3.90
$0.6943 \Rightarrow 0.34715$	53.48	4.76
$1.0642 \Rightarrow 0.5321$	30.08	4.23
$1.3188 \Rightarrow 0.6594$	24.27	3.63
$2.9365 \Rightarrow 1.46825$	20.15	3.04
SFG, $o + o \Rightarrow e$		
$1.0642 + 0.5321 \Rightarrow 0.354733$	47.81	5.00
$1.3188 + 0.6594 \Rightarrow 0.4396$	35.42	4.67

Experimental values of internal angular, temperature and spectral bandwidths ($T = 293\text{K}$):

Interacting wavelengths [μm]	θ_{pm} [deg]	$\Delta\theta^{\text{int}}$ [deg]	ΔT [$^\circ\text{C}$]	$\Delta\nu$ [cm^{-1}]	Ref.
SHG, $o + o \Rightarrow e$					
$0.586 \Rightarrow 0.293$	90	0.5–0.58		2.04	3.195
$0.6943 \Rightarrow 0.34715$	52	0.018			3.197
$1.06 \Rightarrow 0.53$	30	0.019		6.27	3.96
$1.0642 \Rightarrow 0.5321$	30	0.022			3.201
	30	0.022	40		3.216
	30	0.024	52.4		3.217
	30	0.026			3.92
$1.0845 \Rightarrow 0.54225$	29	0.020			3.184

Temperature variation of phase-matching angle:

Interacting wavelengths [μm]	θ_{pm} [deg]	$d\theta_{\text{pm}}/dT$ [deg/K]	Ref.
SHG, $o + o \Rightarrow e$			
$1.0845 \Rightarrow 0.54225$	29	$< -1.3 \times 10^{-3}$	3.184
$1.0642 \Rightarrow 0.5321$	30	-8.4×10^{-4}	3.217

Effective nonlinearity expression in the phase-matching direction:

$$d_{\text{oe}} = d_{31} \sin \theta$$

Nonlinear coefficient [3.37]:

$$d_{31}(1.064 \mu\text{m}) = 4.4 \text{ pm/V} ,$$

$$d_{33}(1.064 \mu\text{m}) = 4.5 \text{ pm/V} .$$

Laser-induced bulk-damage threshold:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m^2]	Ref.	Note
0.44–0.62	200–300	0.1	3.218	
0.53	15	0.4–0.5	3.199	
	20	0.7–0.8	3.219	
0.5321	0.031	50	3.220	
	0.032	100–120	3.221	25 Hz
	0.035	80–100	3.222	1 Hz
	0.035	40–50	3.222	12.5 Hz
	0.1	10	3.220	
	12	0.3	3.202	
0.64	330	0.04	3.101	
0.6943	10	1.2	3.206	
	20	1.3	3.185	10 pulses
	20	0.25	3.203	500 pulses
1.0642	0.045	190	3.220	
	0.13	80	3.220	
	10	1.2	3.223	100 Hz
	12	1.2	3.202	
	300	0.02	3.201	1 kHz
	1.8×10^5	> 0.5	3.224	50 Hz

Thermal conductivity coefficient [3.182]:

$$\kappa = 1.47 \text{ W/mk}$$

3.1.7 KTiOPO₄, Potassium Titanyl Phosphate (KTP)

Positive biaxial crystal: $2V_Z = 37.4^\circ$ at $\lambda = 0.5461 \mu\text{m}$ [3.225];

Point group: mm2

Assignment of dielectric and crystallographic axes:

$X, Y, Z \Rightarrow a, b, c$ (Fig. 3.2);

Mass density: 2.945 g/cm^3 [3.226, 227];

3.023 g/cm^3 [3.228]; 3.03 g/cm^3 [3.229];

Mohs hardness: 5 [3.227];

Vickers hardness: 531 [3.228], 566 [3.230];

Knoop hardness: 702 [3.228];

3.3.28 HgS, CinnibarPositive uniaxial crystal: $n_e > n_o$;

Point group: 32;

Mass density: 8.10 g/cm³ [3.64];

Mohs hardness: 2 – 2.5 [3.64], 3 [3.338];

Transparency range at “0” transmittance level: 0.62 – 13 μm [3.570];Linear absorption coefficient α [3.571]:

λ [μm]	α [cm^{-1}]	Note
0.6328	1.7	o – wave, DFG direction
0.6729	1.4	e – wave, DFG direction
5.3	0.032	o – wave, SHG direction
10.6	0.073	e – wave, SHG and DFG directions

Experimental values of refractive indices [3.570]:

λ [μm]	n_o	n_e	λ [μm]	n_o	n_e
0.62	2.9028	3.2560	2.80	2.6414	2.9052
0.65	2.8655	3.2064	3.00	2.6401	2.9036
0.68	2.8384	3.1703	3.20	2.6387	2.9017
0.70	2.8224	3.1489	3.40	2.6375	2.9001
0.80	2.7704	3.0743	3.60	2.6358	2.8987
0.90	2.7383	3.0340	3.80	2.6353	2.8971
1.00	2.7120	3.0050	4.00	2.6348	2.8963
1.20	2.6884	2.9680	5.00	2.6267	2.8863
1.40	2.6730	2.9475	6.00	2.6233	2.8799
1.60	2.6633	2.9344	7.00	2.6156	2.8741
1.80	2.6567	2.9258	8.00	2.6112	2.8674
2.00	2.6518	2.9194	9.00	2.6066	2.8608
2.20	2.6483	2.9146	10.00	2.6018	2.8522
2.40	2.6455	2.9108	11.00	2.5914	2.8434
2.60	2.6433	2.9079			

Optical activity [3.194]:

λ [μm]	ρ [deg/mm]	λ [μm]	ρ [deg/mm]
0.6058	447	0.7281	145
0.6131	393.5	0.7789	113.5
0.6278	319	0.8296	92.5
0.6424	270.5	0.8757	74.5
0.6571	237.5	0.9196	65.5
0.6681	218	0.9527	59
0.6770	200	0.9967	51.5

Experimental values of phase-matching angle ($T = 293$ K) and comparison between different sets of dispersion relations:

Interacting wavelengths [μm]	θ_{exp} [deg]	θ_{theor} [deg]		
		[3.458]	[3.362]	[3.543]
SHG, $e + e \Rightarrow o$				
10.6 $\Rightarrow$ 5.3	20.8 [3.571] 21.5 [3.572]	21.2	23.0	21.3
SFG, $e + e \Rightarrow o$				
10.6 + 0.6729 $\Rightarrow$ 0.6328	25.3 [3.571]	no pm	25.8	25.7

Best set of dispersion relations (λ in μm , $T = 20$ °C) [3.543]:

$$n_o^2 = 7.8113 + \frac{0.3944}{\lambda^2 - 0.1172} + \frac{604.5}{\lambda^2 - 682.5} ,$$

$$n_e^2 = 9.3139 + \frac{0.5870}{\lambda^2 - 0.1166} + \frac{542.6}{\lambda^2 - 540.8} .$$

Calculated values of phase-matching and “walk-off” angles:

Interacting wavelengths [μm]	θ_{pm} [deg]	ρ_1 [deg]	ρ_2 [deg]
SHG, $e + e \Rightarrow o$			
$10.6 \Rightarrow 5.3$	21.32	3.19	3.19
$9.6 \Rightarrow 4.8$	19.09	2.93	2.93
$5.3 \Rightarrow 2.65$	14.42	2.32	2.32
$4.8 \Rightarrow 2.4$	14.82	2.38	2.38
$2.65 \Rightarrow 1.325$	23.44	3.61	3.61
$2.4 \Rightarrow 1.2$	26.00	3.93	3.93
SHG, $o + e \Rightarrow o$			
$10.6 \Rightarrow 5.3$	30.68		4.22
$9.6 \Rightarrow 4.8$	27.36		3.93
$5.3 \Rightarrow 2.65$	20.54		3.19
$4.8 \Rightarrow 2.4$	21.12		3.27
$2.65 \Rightarrow 1.325$	33.85		4.69
$2.4 \Rightarrow 1.2$	37.78		4.98
SFG, $o + e \Rightarrow o$			
$10.6 + 5.3 \Rightarrow 3.533$	19.75		3.08
$9.6 + 4.8 \Rightarrow 3.2$	18.65		2.94
SFG, $e + o \Rightarrow o$			
$10.6 + 5.3 \Rightarrow 3.533$	28.96	4.18	
$9.6 + 2.4 \Rightarrow 1.92$	27.16	4.00	

Effective nonlinearity expressions in the phase-matching direction [3.100]:

$$d_{\text{eeo}} = d_{11} \cos^2 \theta \sin 3\phi ,$$

$$d_{\text{oeo}} = d_{\text{eoo}} = d_{11} \cos \theta \cos 3\phi .$$

Nonlinear coefficient:

$$d_{11}(10.6 \mu\text{m}) = 50 \pm 16 \text{ pm/V [3.365]} :$$

Laser-induced surface-damage threshold [3.365]:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m^2]
1.06	17	0.4

3.3.29 Ag_3SbS_3 , Pyrargyrite

Negative uniaxial crystal: $n_o > n_e$;

Point group: 3m;

Mass density: $5.83 \text{ g}/\text{cm}^3$ [3.64];

Mohs hardness: 2 – 2.5 [3.64];

Transparency range at “0” transmittance level: $0.7\text{--}14 \mu\text{m}$ [3.573];

Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
0.967	≈ 0.7	3.574	e – wave, SFG direction
1.064	≈ 0.7	3.574	o – wave, SFG direction
10.6	≈ 0.7	3.574	o – wave, SFG direction
	0.5	3.455	o – wave, SHG direction
	0.34	3.575	o – wave, $\parallel c$
	0.08	3.575	e – wave, $\perp c$
13.5	< 1	3.573	

The graph of n_o and n_e dependences versus wavelength is given in [3.573].

Sellmeier equations (λ in μm , $T = 20^\circ\text{C}$) [3.573]:

$$n_o^2 = 1 + \frac{6.585 \lambda^2}{\lambda^2 - 0.16} + \frac{0.1133 \lambda^2}{\lambda^2 - 225} ,$$

$$n_e^2 = 1 + \frac{5.845 \lambda^2}{\lambda^2 - 0.16} + \frac{0.0202 \lambda^2}{\lambda^2 - 225} .$$

Experimental and theoretical values of phase-matching angle and calculated values of “walk-off” angle:

Interacting wavelengths [μm]	θ_{exp} [deg]	θ_{theor} [deg] [3.573]	ρ_3 [deg] [3.573]
SHG, $o + o \Rightarrow e$			
$10.59 \Rightarrow 5.295$	30 [3.455]	23.34	2.18
$10.6 \Rightarrow 5.3$	29 [3.575]	23.37	2.18
	27.6 [3.574]		

SFG, $o + o \Rightarrow e$

$$10.6 + 1.064 \Rightarrow 0.967 \quad 27.3 \text{ [3.574]} \quad 39.65 \quad 2.98$$

Effective nonlinearity expressions in the phase-matching direction [3.100]:

$$d_{oeo} = d_{31} \sin \theta - d_{22} \cos \theta \sin 3\phi ,$$

$$d_{eoe} = d_{oeo} = d_{22} \cos^2 \theta \cos 3\phi .$$

Nonlinear coefficients:

$$d_+(10.6 \mu\text{m}) = (7.5 \pm 0.3)^{-1} \times |d_{36}(\text{GaAs})|$$

$$= 11.1 \pm 0.4 \text{ pm/V [3.576, 37].}$$

Using the ratio

$$|d_{22}(\text{Ag}_3\text{SbS}_3)| / |d_{31}(\text{Ag}_3\text{SbS}_3)| = 1.05 \pm 0.04 \text{ [3.575]}$$

and the values

$$\theta_{\text{pm}}(10.6 \Rightarrow 5.3 \mu\text{m}, o + o \Rightarrow e) = 29^\circ, \rho = 2.4^\circ \text{ [3.575]}$$

from the equation

$$d_+ = |d_{31}| \sin(\theta_{\text{pm}} + \rho) + |d_{22}| \cos(\theta_{\text{pm}} + \rho)$$

we deduce

$$|d_{22}(10.6 \mu\text{m})| = 8.2 \pm 0.8 \text{ pm/V} ,$$

$$|d_{31}(10.6 \mu\text{m})| = 7.8 \pm 0.5 \text{ pm/V}.$$

Laser-induced surface-damage threshold [3.365]:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m^2]
1.06	17	>0.09
10.6	200	>0.46

3.3.30 Se, Selenium

Positive uniaxial crystal: $n_e > n_o$;

Point group: 32;

Mass density: $4.79 \text{ g}/\text{cm}^3$ [3.59];

Mohs hardness: 2 [3.59];

Transparency range at “0” transmittance level: $0.7 - 21 \mu\text{m}$ [3.577, 578];Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
5.3	1.40 ± 0.05	3.579	$\parallel c$
10.6	1.09 ± 0.02	3.579	$\parallel c$

λ [μm]	α [cm^{-1}]	Ref.	Note
14	2.8 ± 0.5	3.580	o – wave, $\perp c$
28	50 ± 5	3.580	o – wave, $\perp c$

Experimental values of refractive indices at 296 K [3.581]:

λ [μm]	n_o	n_e
1.064	1.790 ± 0.008	3.608 ± 0.008
1.1523	2.737 ± 0.008	3.573 ± 0.008
3.3913	2.650 ± 0.01	3.460 ± 0.01
10.6	2.640 ± 0.01	3.410 ± 0.01

Optical activity:

λ [μm]	ρ [deg/mm]	Ref.
0.70	440 ± 20	3.582
0.79	300 ± 15	3.582
0.91	200 ± 15	3.582
1.00	150 ± 10	3.582
1.14	100 ± 10	3.582
3.39	4.8 ± 0.5	3.579
10.6	2.5 ± 0.5	3.579

Experimental values of phase-matching angle:

Interacting wavelengths [μm]	θ_{pm} [deg]
SHG, $e + e \Rightarrow o$	
$10.6 \Rightarrow 5.3$	5.5 ± 0.3 [3.579] 6.5 [3.577] ≈ 10 [3.583]

Effective nonlinearity expressions in phase-matching direction [3.100]:

$$d_{\text{eoo}} = d_{11} \cos^2 \theta \sin 3\phi$$

$$d_{\text{ooo}} = d_{\text{eoo}} = d_{11} \cos \theta \cos 3\phi$$

Nonlinear coefficient:

$$d_{11}(10.6 \mu\text{m}) = 97 \pm 25 \text{ pm/V} [3.579]$$

Thermal conductivity coefficient [3.584]:

T [K]	κ [W/mK], $\parallel c$	κ [W/mK], $\perp c$
273	4.81	1.37
298	4.52	1.31

3.3.31 Ti_3AsS_3 , Thallium Arsenic Selenide (TAS)

Negative uniaxial crystal: $n_o > n_e$;

Point group: 3m;

Mass density: 7.83 [3.585];

Mohs hardness: 2 – 3 [3.586];

Transparency range at 0.5 transmittance level for a 6 mm long crystal:

1.28 – 17 μm [3.586];

Linear absorption coefficient α :

λ [μm]	α [cm^{-1}]	Ref.	Note
2–12	< 0.02	3.585	
10.6	0.082	3.454	SHG direction
	0.038	3.586	

Experimental values of refractive indices at 300 K [3.587]:

λ [μm]	n_o	n_e
2.056	3.419	3.227
3.059	3.380	3.190
4.060	3.364	3.177
5.035	3.357	3.171
5.856	3.354	3.168
6.945	3.349	3.164
7.854	3.345	3.162
9.016	3.340	3.158
9.917	3.336	3.155
10.961	3.331	3.152
12.028	3.327	3.147

Temperature derivative of refractive indices at

$\lambda = 2 - 10.6 \mu\text{m}$ ($T = 80 - 300 \text{ K}$) [3.587] :

$$\frac{dn_o}{dT} = -4.52 \times 10^{-5} \text{ K}^{-1} ;$$

$$\frac{dn_e}{dT} = +3.55 \times 10^{-5} \text{ K}^{-1} .$$

Sellmeier equations (λ in μm , $T = 27^\circ\text{C}$) [3.587]:

$$n_o^2 = 1 + \frac{10.210 \lambda^2}{\lambda^2 - 0.197136} + \frac{0.522 \lambda^2}{\lambda^2 - 625} ,$$

$$n_e^2 = 1 + \frac{8.993 \lambda^2}{\lambda^2 - 0.197136} + \frac{0.308 \lambda^2}{\lambda^2 - 625} .$$

Calculated values of phase-matching and “walk-off” angles:

Interacting wavelengths [μm]	θ_{pm} [deg]	ρ_1 [deg]	ρ_2 [deg]	ρ_3 [deg]
SHG, o + o $\Rightarrow$ e				
10.6 $\Rightarrow$ 5.3	19.10			2.12
9.6 $\Rightarrow$ 4.8	18.54			2.07
5.3 $\Rightarrow$ 2.65	24.79			2.60
4.8 $\Rightarrow$ 2.4	27.26			2.77
2.9365 $\Rightarrow$ 1.46825	48.74			3.26
SFG, o + o $\Rightarrow$ e				
10.6 + 2.65 $\Rightarrow$ 2.12	25.21			2.64
9.6 + 2.4 $\Rightarrow$ 1.92	27.65			2.81
SHG, e + o $\Rightarrow$ e				
10.6 $\Rightarrow$ 5.3	26.79	2.65		2.72
9.6 $\Rightarrow$ 4.8	26.03	2.62		2.67
5.3 $\Rightarrow$ 2.65	35.77	3.16		3.18
4.8 $\Rightarrow$ 2.4	39.78	3.25		3.27
SFG, e + o $\Rightarrow$ e				
10.6 + 5.3 $\Rightarrow$ 3.533	23.81	2.45		2.52
9.6 + 4.8 $\Rightarrow$ 3.2	25.06	2.55		2.62
SFG, o + e $\Rightarrow$ e				
10.6 + 5.3 $\Rightarrow$ 3.533	34.84		3.13	3.14
9.6 + 4.8 $\Rightarrow$ 3.2	36.84		3.19	3.20

Experimental values of internal angular bandwidth:

Interacting wavelengths [μm]	$\Delta\theta^{\text{int}}$ [deg]	Ref.
SHG, o + o $\Rightarrow$ e		
9.6 $\Rightarrow$ 4.8	0.27	3.588
10.6 $\Rightarrow$ 5.3	0.30	3.589

Effective nonlinearity expressions in the phase-matching direction [3.100]:

$$d_{\text{ooe}} = d_{31} \sin \theta - d_{22} \cos \theta \sin 3\phi ,$$

$$d_{\text{eoe}} = d_{\text{oeo}} = d_{22} \cos^2 \theta \cos 3\phi .$$

Nonlinear coefficient:

$$\begin{aligned} d_+(10.6 \mu\text{m}) &= (3.47 \pm 1.04) \times d_+(\text{Ag}_3\text{AsS}_3) \\ &= 67.5 \pm 31.3 \text{ pm/V [3.586, 455, 37]} , \end{aligned}$$

$$\begin{aligned} d_+(10.6 \mu\text{m}) &= (3.3 \pm 1.0) \times d_+(\text{Ag}_3\text{SbS}_3) \\ &= 36.5 \pm 12.5 \text{ pm/V [3.586, 576, 37]} . \end{aligned}$$

Laser-induced surface-damage threshold:

λ [μm]	τ_p [ns]	$I_{\text{thr}} \times 10^{-12}$ [W/m ²]	Ref.
9.6	70	> 0.054	3.588
10.6	150	0.1-0.17	3.368
10.6	200	0.16	3.586

3.3.32 Te, Tellurium

Positive uniaxial crystal: $n_e > n_o$;

Point group: 32;

Mass density: 6.25 g/cm³ [3.59];

Mohs hardness: 2 – 2.5 [3.59];

Transparency range at “0” transmittance level: 3.5 – 36 μm [3.590, 578, 591];

Linear absorption coefficient α :

λ [μm]	α [cm ⁻¹]	Ref.	Note
5.3	1.32	3.451	o – wave, SHG direction
10.6	0.96	3.451	e – wave, SHG direction
	0.5–1.0	3.576	e – wave, SHG direction
	0.2–0.6	3.592	e – wave, SHG direction
14	1.1 $\pm$ 0.4	3.580	o – wave, $\perp c$
28	4.4 $\pm$ 0.04	3.580	o – wave, $\perp c$

Two-photon absorption coefficient β [3.593]:

Wavelengths of absorbed photons [μm]	$\beta \times 10^9$ [m/W]
5.3 + 5.3	8
5.3 + 10.6	2

Experimental values of refractive indices:

λ [μm]	n_o	n_e	Ref.	λ [μm]	n_o	n_e	Ref.
4.0	4.929	6.372	3.578	10.6	4.792	6.247	3.590
5.0	4.864	6.316	3.578	10.8	4.791	6.246	3.590
6.0	4.838	6.286	3.578	11.4	4.789	6.243	3.590
7.0	4.821	6.257	3.578	12.0	4.785	6.240	3.590
8.0	4.809	6.253	3.578	12.8	4.781	6.235	3.590
8.5	4.801	6.260	3.590	13.7	4.776	6.231	3.590
8.8	4.799	6.258	3.590	14.0	4.775	6.230	3.590
9.3	4.798	6.255	3.590	14.7	4.772	6.227	3.590
9.7	4.795	6.252	3.590	15.9	4.767	6.222	3.590
10.2	4.793	6.249	3.590	17.2	4.761	6.216	3.590